

1.3 Neuro-muscular system	1.3.1	Knowledge, understanding and application of the anatomy and physiology and the function of the neuro-muscular system during physical activity.
	1.3.2	The characteristics and anatomical make-up of the different fibre types: slow twitch (type I), fast oxidative glycolytic (IIa) and fast glycolytic (type IIX, formerly known as IIB).
	1.3.3	The different structure of each fibre type: how it facilitates their physiology and affects their suitability for particular types of physical activities.
	1.3.4	The fibre recruitment patterns for endurance and power-based events, and how specific training can enable athletes to gain control over the recruitment pattern.
	1.3.5	The anatomy of the neuro-muscular system, including the central nervous system, muscle fibres, myofibrils, sarcomeres, motor units, motor neurones and neuro-muscular end plates, the protein filaments of actin and myosin, and the roles of the globular proteins of troponin and tropomyosin.
	1.3.6	The physiology of a muscular contraction, from a nervous impulse to a muscular response. To include: the neuro-muscular transfer, sliding filament theory, the all-or-none law. Knowledge of the five stages of a muscle contraction (resting, excitation, contraction, recharge and relaxing). Understanding of wave summation and gradation of contraction.
	1.3.7	Understanding of how the neuro-muscular system responds acutely, both structurally and functionally, to the stress of warming up and immediate physical or sporting activity.
	1.3.8	The chronic adaptations of the cardiorespiratory, cardiovascular, muscular-skeletal and neuro-muscular systems to training.